



How We Tested

The testing of speakers was completely subjective. Our test-bed consisted of an Intel 875PBZ motherboard running a Pentium 4 3 GHz with 512 MB 333 MHz DDR RAM. The sound card used was the SoundBlaster Audigy2 ZS Platinum PRO that supports 7.1 speakers, while the video card used was the Radeon 9600 PRO.

Our tests involved two steps—first, we kept the sound control volume—bass, treble, master, wave, speaker master, etc.—at 50 per cent and ran the tests to check the quality of the sound output. Then we increased the volume to the maximum for both, sound card and speakers, and carried out the same tests. We used the native sound control panel for the sound card, and the volume control on the speakers. Equalizers were never turned on, to avoid any effects.

Features, Ease of Use, and Manuals and Documentation were assigned 15 per cent, 10 per cent and 5 per cent of the total scores respectively. The most important, performance, earned 70 per cent.

We carried out tests that involved everyday media plus some special frequency tests. Ratings were given, on a scale of 5, for each test by three different reviewers, and then we calculated an average score to reduce subjectivity. Our tests involved the following benchmarks.

Gaming tests: We carried out two gaming benchmarks—*Aquamark3* and *Quake III*—and the sound quality for each was noted. We ran the complete *Aquamark3* demo and the *Quake III* Intro movie.

Movie test: This involved DVD playback of the *Minority Report* automobile assembly fight sequence. This scene is noted for low frequency effects of the shockwave gun, and the general atmosphere involving metallic objects and lasers in the plant, which have a sharp tone.

Music tests: Three separate tests were run here—an Audio CD test, MP3 playback and DVD audio—comprising both Hindi and English songs. The songs we chose for the Audio CD test were 'With or Without You' and 'Bullet the Blue Sky' by U2 from the album *The Joshua Tree*, 'O Rey Chhori' from *Lagaan* and 'Rachna Teen Taal' from Ustad Amjad Ali Khan's album *Raag Yaman*. The MP3s we played were 'Hotel California' by the Eagles and 'Duur' (Reprise) by Strings. For the DVD audio test, we played *Shut Up* by Vanessa Lausdei.

Each Audio CD song was chosen for a reason: *With or Without You* has deep percussive bass that test the subwoofer; *Bullet the Blue Sky* has guitar fireworks that test the mids and highs at full volume; *O Rey Chhori* tests the reproduction of vocals while the sprinkling of piano pieces simultaneously tests the highs; *Rachna Teen Taal* tests all the speakers with the *Santoor* and *Tabla* combination.

MP3s are compressed audio that generally sound good on any speaker. We used the normal 128 Kbps, joint-stereo format. *Hotel California* tested the subwoofer with its deep bass notes, and *Duur* checked the treble reproduction.

The DVD audio song is encoded in five discrete channels, and features excellent mid-range, treble and vocals.

Frequency Tests: These tests involved the playback of special wav files corresponding to the frequencies of 50 Hz, 100 Hz, 250 Hz, 500 Hz (left and right channels), 1 KHz, 10 KHz and 15 KHz. Points were given to the speakers the could reproduce the tones perfectly.

Overall Bass and Treble Tests: We played an audio file to check the overall bass reproduction, and did the same to check the treble.

Point One Out

Four point one, Five point one... stuck in decimals? Here's a small primer on the different surround formats out there:

4/4.1 format: This is a psuedo-surround format as it simulates the surround effect to a certain degree. It utilises two channels—the left and right stereo channels. The surround channel information for the rear speakers is encoded on to the original stereo channels at a set phase difference and then supplied to the rear speakers. In case of a separate sub-woofer, all bass effects are diverted to it via a suitable crossover that's located on the sub-woofer itself or the amplifier. This format is on its way out, especially with the falling prices of 5.1 and higher certified audio systems.

5.1 surround: Also known as Dolby Digital, this surround-sound format was introduced by Dolby Systems and was first utilised in the soundtrack of *Batman* in 1992. It was the first to feature a dedicated centre channel, as well as a low frequency effects (LFE) channel, in addition to the four existing channels. Also, this was the first true surround format that had truly discrete channels, i.e., all the speakers received an independent signal without any distortion or channel mixing. Dolby Digital was followed by

Digital Theatre System (DTS), which offered better compression than Dolby Digital, giving better audio quality. Today, most movies and many games are encoded using either the Dolby Digital or DTS formats. The centre channel here assumes a good deal of importance, as all human voices are channelled through it to provide a centralised listening experience.

6.1 surround: In this surround setup, an additional rear centre speaker is connected to the existing five plus one setup. The signal for this speaker is obtained by separating a central rear channel from the original left and right rear signals. The Dolby Digital EX and DTS Pro Neo 6 can output 6.1 channel signals. As this is a comparatively newer format, there are only a limited number of movie soundtracks that feature this.

7.1 surround: This is a top-of-the-line setup, with three speakers in the front, including the centre unit, two speakers on each side of the listener, and two rear speakers to complete the surround effect. Currently, there are no applications or movies that can utilise such a setup, but some decoders have the ability to upmix 5.1 or 6.1 sound to give a 7.1 output.